

Primer-Derived Research Program: Residual Work Register

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Compiled by Codex

A residual-only execution register after reconciling the six primer backlogs with merged contracts, experiments, M5, and the locked root quality gate. Completed negative results are preserved as completions; invalid runs remain non-evidence; overlapping work is collapsed into dependency-ordered waves.

Baseline	origin/main 0413098
Reconciled	2026-07-14
Scope	Six primers and shared program infrastructure
Disposition	Residual work only

This artifact is a planning synthesis. Canonical scientific state remains in experiment manifests, evidence records, gate verdicts, and preregistrations.

Post-merge reconciliation against origin/main commit 0413098 on 2026-07-14.

Human research director: Jawaun Brown. Compiled by Codex from the six primer backlogs, merged experiment artifacts, the active research ledger, and the discovery-regime audit.

Executive conclusion

The primer program is no longer a greenfield plan. It already has a structured evidence and claim kernel, five experiment manifests, one canonical gate-verdict migration, six corrected primers, an exact finite Bayesian value-of-information benchmark, a seed/bootstrap calibration study, a passive-active coupling and hysteresis study, a reusable synthetic causal-use API, a locked root quality environment, and the completed M5 reopen-trigger comparison.

M5 is complete as an experiment even though its universal superiority claim failed. F0, F1, F4, and F5 passed; F2 failed because the periodic trigger tied commitment latency, and F3 failed because a never-firing normalized trigger tied the zero false-reopen rate. The surviving claim is only that commitment change gave the best observed Pareto combination in this finite Suite C schedule. M5 is search inside an existing regime, not a discovery-regime transition or a neural continual-learning result.

The residual program is therefore narrower than the original six backlogs. It has four dominant bottlenecks:

- Finish the trust substrate: migrate structured gates, claims, manifests, status transitions, preregistration chronology, and provenance across the program.
- Make flagship evidence independently recomputable: public-safe rows, exact lineage, clean-clone verification, and outside replication bundles.
- Replace task-local positive stories with decisive mechanism tests: theorem assumptions, endogenous concern, graded causal use, dynamical return, viability, and many-part coordination.
- Add independence and program-level statistics: multiplicity, sequential stopping, adversarial control design, blinded claim-tier review, and external promotion criteria.

How to read this register

- `complete` means the requested artifact or experiment exists. A strict FAIL or negative result can still complete an experiment.
- `stronger partial` means merged work materially narrows the residual but does not satisfy the original completion gate.
- `open` means the decisive deliverable has not been built or run.
- `invalid` means execution occurred but the frozen integrity gate blocked scientific interpretation; it must not be counted as a positive or negative verdict.
- IDs are preserved from the article-specific backlogs. Completed IDs are omitted from the residual lists unless needed to prevent accidental reruns.

This document optimizes overlap. The article sections preserve source ownership; the final execution waves combine duplicate work into one program sequence.

Work that is no longer residual

- **M5 / H-209 - complete experiment, strict FAIL.** Five trigger arms, eight paired seeds, exact probe budgets, coupled no-change controls, paired bootstrap intervals, preregistration, calibration receipt, branch-independent random variates, cross-platform semantic hashes, and a frozen integrity manifest are implemented. Do not tune M5 or relist it as unrun.
- **M-208 - complete bounded synthetic experiment.** Exact finite Bayesian value of information is implemented across learnable, irreducible-noise, and misspecified regimes. High information can have zero oracle value, and the heuristic cannot rank probes in the registered counterexample.
- **T-SYS-011 and T-SYS-012 - complete bounded synthetic tranches.** The bifurcation gate is rejected; matched-budget hysteresis and path dependence pass. This is not evidence for a general phase transition or autonomous attractor.
- **P-01, P-02, P-03, P-05, P-06, P-07, P-08, and P-28 - complete at their stated primer or synthetic-API scopes.** Identity metadata, claim-tier matrix, three concern senses, corrected evidence history, model-relative null anchoring, and graded causal-use utilities are present.
- **S-001 - complete.** Science-primer identity metadata is corrected and checked.
- **E-SE-001 - complete.** Software-primer identity metadata is corrected and included in the six-primer metadata gate.
- **Root quality gate - implemented foundation.** `uv.lock`, a PEP 735 quality group, CPU-only Torch resolution, all 70 root tests, bounded xdist, contract validators, metadata checks, provenance freshness, Ruff, and `ty` run in required PR CI. The residual is cross-project coverage and adversarial validation, not rebuilding this root lane.

Dependency-ordered execution map

Wave 0 - Repair the live ledger and article truth

Update every stale status before new experimentation. Mark H-209 complete with a strict negative verdict; replace the history primer's "M5 unrun" language; update the software primer from the old unpinned `uvx` and absent-CI model to the locked root-quality reality; propagate corrected mathematics and systems language into downstream papers. Generate status and inventory fragments so the same drift fails CI.

Primary IDs: H-001, H-005, M-001-M-010, M-012-M-014, P-04, P-24, P-29, S-002-S-008, E-SE-002-E-SE-009, E-SE-037, T-SYS-001-T-SYS-004.

Wave 1 - Finish the research-instrument kernel

Promote the existing contracts into a program-wide kernel: one versioned manifest, evidence record, claim record, gate verdict, metric identifier, resampling unit, status transition, preregistration amendment, public-artifact envelope, and execution lineage per claim-bearing run. Add legacy exceptions with owners and expiry rather than silently accepting uncovered packages.

Primary IDs: H-401-H-403, M-401-M-408, P-26, P-27, P-29, S-036-S-048, E-SE-023-E-SE-042, T-SYS-028-T-SYS-031.

Wave 2 - Establish program-level statistical control

Audit every three-seed promotion, finish empirical seed-floor calibration, encode hierarchical sampling units, add selective-inference and family-level error policies, and preregister one prospective claim family with stopping and multiplicity rules. Do not use a universal seed count.

Primary IDs: M-004, M-103, M-205, M-214, M-403, S-006, S-009, S-022, S-030, S-037.

Wave 3 - Make evidence recomputable and independently auditable

Publish the minimum safe rows needed to recompute flagship statistics, bind every output to command, commit, environment, model, data, and parent digests, turn `regen.py` into a verifier, and create clean-room audit bundles. Run the first second-host clean-clone drill only after these contracts exist.

Primary IDs: H-006, H-403, P-25, S-003, S-018-S-021, S-029, S-039, S-040, S-045, S-046, S-049, S-050, E-SE-010-E-SE-015, E-SE-019-E-SE-021, E-SE-025, E-SE-030, E-SE-033, E-SE-036, E-SE-039, E-SE-040.

Wave 4 - Correct old experiments before expanding claims

Repair known estimator, provenance, control, and function-family failures. Preserve kills and invalid runs. Do not spend on a larger model when the existing claim failed for an internal reason that the new scale does not address.

Primary IDs: H-101-H-109, M-101-M-111, P-09, P-11, P-14, P-17, S-010-S-017, T-SYS-005-T-SYS-010.

Wave 5 - Run the decisive core mechanism tests

Test theorem necessity, endogenous concern, unrecoverable self structure, real-model and multi-surface causal-use dose response, topology/spectrum/weakness dissociation, return dynamics, viability kernels, strict self-production, and maintenance cost. Each experiment must include a strongest impostor and a negative control designed to pass behavior while failing mechanism.

Primary IDs: H-201-H-210 except H-209, M-201-M-216 except M-208, P-10, P-12, P-13, P-15, P-30, P-31, S-023-S-026, T-SYS-013, T-SYS-017-T-SYS-019, T-SYS-034.

Wave 6 - Expand only through external and many-part contact

After the evidence substrate is inspectable, move to outsider-authored environments, independent adversaries, natural images and open semantics, distributed coordination, collective navigation, evolving populations, and biological or multiscale contact with operational matches.

Primary IDs: H-101-H-103, H-205, H-503, M-105, M-110, M-203, M-210, M-211, M-215, M-216, P-13, P-16, P-25, S-018-S-021, S-025, S-028, S-049, S-050, T-SYS-014-T-SYS-016, T-SYS-029, T-SYS-032.

Wave 7 - Complete reading only where it changes a gate

Finish the named philosophy, science, mathematics, systems, history, and software reading packets. Every note must terminate in a claim boundary, control, estimator, schema field, or decision rule. Reading without an operational consequence remains background, not a completion.

Article 1 - History, lineage, and trajectory residuals

Article correction residuals

- **H-001 - stronger partial, reopened by new evidence.** The primer was corrected once, but it again says M5 is unrun. Refresh the dated frontier, record M5's strict FAIL, retain E7 as integrity-invalid/no-verdict, and regenerate the PDF. Completion gate: primer, backlog, active ledger, and PDF all agree with the latest canonical result records.
- **H-002 and H-003 - open.** Replace a single convergence story with competing diffusion, independent-convergence, and shared-tooling hypotheses; repair the intellectual attribution graph using primary sources. Completion gate: each historical hypothesis has discriminating observations and each lineage edge has a source.
- **H-005 - stronger partial.** Candidate-law and novelty language is bounded in the primers, but the downstream paper audit is incomplete. Completion gate: all abstracts and conclusions use the evidence-tier vocabulary and external kills cannot coexist with universal-law wording.
- **H-006 - open.** Restore missing external-contact primary reports and provenance rather than citing handoff summaries. Completion gate: every reported external number resolves to a tracked result artifact and provenance card.

Old experiments to correct or replicate

- **H-101 - open with a preserved kill.** Treat Pythia P1 as an accepted negative. Run again only in a genuinely different external regime with a preregistered reason the old failure may not transport.
- **H-102 and H-103 - open.** Define explicit invariant types for cross-substrate geometry; complete the grid-cell negative with topology robustness and external recordings. Completion gate: similarity is typed as linear, kernel, topological, dynamical, or interventional before analysis.
- **H-104 - open.** Replace the inferred effective-dimension slope with dimension-conditioned measurement and estimator checks.
- **H-105 and H-106 - open.** Repair label-dependent/body-level controls and consolidate the probing correction chain across worlds.
- **H-107-H-109 - open.** Resolve the role-specific architecture ceiling, reconcile uncertainty and concern geometry by mechanism, and treat the sharp boundary as a function-family study rather than a universal discontinuity.

New experiment residuals

- **H-201-H-205 - open.** Test historical independence versus diffusion; causally manipulate capacity, concern, and shared statistics; build strict minimal autopoiesis; compute viability kernels; and move intervention invention to natural images and open semantics.
- **H-206 - open and requires a new preregistration.** Redesign E6 around reward-density feasibility. The blocked smoke cannot be repaired post hoc under the old confirmatory contract.
- **H-207 - open.** Run E5-L longitudinal generator-versus-coverage separation only after calibration and cost approval.
- **H-208 - invalid current run, new preregistration required.** Preserve the 32-stream timing-invalid artifact as non-evidence. Any replacement needs a non-tautological budget verifier frozen before outcomes.

- **H-210 - open.** Put compatibility, concern weighting, and action coupling in one factorial system and demand preregistered mediation or publish their dissociation.

Reading, framework, and direction residuals

- **H-301-H-304 - open reading packets.** Primary cybernetics/autopoiesis/enactivism; teleosemantics and causal-role kinds; strong alternatives to weakness; viability theory versus active inference.
- **H-401 - open.** The proxy-resistant experiment-package skill is not yet implemented.
- **H-402 - stronger partial.** Claim/evidence registries and validators exist; most experiments and claims remain unmigrated.
- **H-403 - open.** Build external-replication adapters and provenance checks.
- **H-501-H-503 - open directions.** Prefer a regime map over a universal weakness scalar; approach biological/multiscale agency only through operational matches; require independent replication for promotion of machine-made science.

Article 2 - Mathematics of constraint residuals

Article correction residuals

- **M-001 and M-002 - stronger partial, propagation unfinished.** The primer and numeric tests are corrected, but downstream notes and papers still contain the negative-price Lagrangian and reversed error-reduction sign. Completion gate: repository-wide claim search is clean and executable examples pin every corrected formula.
- **M-003-M-010 - open or partial truth repair.** Replace stale topology mediation; add statistical assumption boxes; repair geometry/spectral definitions; separate standard equivariance from compatibility count; correct the torus derivation; bound intervention and gauge claims; separate stability, repair, ultrastability, and autopoiesis; add a live claim-status layer.
- **M-011-M-015 - open publication repair.** Add primary chapter references, improve pagination/accessibility, normalize notation and edge cases, rename baseline claims, explain flatness non-identifiability, and add assumption-exposing exercises.

Old experiment residuals

- **M-101-M-104 - open.** Re-derive reward deformation with estimator gates; stress the negative topology mediation result; reanalyze weakness with stronger/dependence-aware baselines; repair learned-group controls and enforce coherence.
- **M-105-M-108 - open.** Replicate behavioral OOD language evidence, predictive policy closure, distributed reward subspaces, and null-action gauge fixing under interaction and drift.
- **M-109-M-111 - open.** Test autonomous repair and the quantitative Stack bound, export gauge-transport gates beyond synthetic L4, and re-evaluate discontinuity under the actual visitation measure.

New experiment residuals

- **M-201 - stronger partial.** Assumption cards and counterexample predicates exist; theorem-conclusion necessity and minimality remain unproved. Completion gate: each claimed theorem has satisfying witnesses, falsifying assumption removals, and a checked conclusion.
- **M-202-M-207 - open.** Weakness regret under misspecification; non-synthetic concern-weighted weakness; causal topology/spectrum/weakness/ODD separation; calibrated persistent homology; manipulated intrinsic dimension; uncertain long-horizon policy closure.
- **M-209-M-213 - open.** Locate boundary failure; learn generators without finite enumeration; build partial-orbit deployment; test reparameterization invariance; validate activation patching as causal abstraction.
- **M-214-M-216 - open.** Test a PAC-Bayes connection, measure weakness in activation spaces, and define same geometry across substrates with explicit invariant types.

Reading, framework, and direction residuals

- **M-301-M-310 - open reading program.** Constrained optimization/quantization; Riemannian geometry; small-sample uncertainty; flatness/PAC-Bayes/MDL; group representations; TDA/manifold inference; causal neural interventions; Bayesian design/control; domain shift; autopoiesis/ultrastability.
- **M-401 and M-402 - stronger partial.** Contracts, templates, validators, selected manifests, and a selected mathematical claim registry exist. The repo-local framework skill, scaffolder, and equation-complete migration do not.

- **M-403-M-406 and M-408 - open.** Dependence-aware statistics toolkit, canonical geometry/topology metrics, transformation/action validators, primer lint/render QA, and reusable causal-intervention harness.
- **M-407 - stronger partial.** M5 demonstrates careful supersession and status synchronization, but prose and provenance are not generated universally from canonical records.
- **M-501-M-506 - open directions.** Estimate concern distributions; separate diagnostic/predictive/causal geometry; test the six-equation unification; treat topology as footprint; replace post-hoc dimension with optimization dynamics; audit discovery versus accumulating task-local probes.

Article 3 - Philosophy: what it means residuals

Article and conceptual residuals

- **P-04 - stronger partial.** The primer has a consciousness/normativity firewall; downstream papers and public summaries are not uniformly checked. Completion gate: no computational result is promoted to consciousness or moral status without an explicit independent argument.
- **P-09 - open.** Build an operating-regime map for weakness rather than a universal scalar story.
- **P-10 - open, central experiment.** Replicate endogenous concern against yoked and wrong-value controls. Completion gate: the learned stake variable predicts and causally affects commitment better than matched exogenous proxies without label leakage.
- **P-11 and P-12 - open.** Decode discarded sensory information; construct an interventionally unrecoverable self rather than a merely hard-to-read one.
- **P-13 - open beyond the synthetic API.** Run graded causal-use dose response across real models, widths, seeds, and at least two commitment surfaces.
- **P-14 - open.** Re-measure tapestry geometry inside the computation that actually uses it.
- **P-15 - stronger partial.** The synthetic causal-use fixture and M5 both instantiate strong impostors; the requested four-suite benchmark and aggregate adversarial scorecard are absent.
- **P-16 - open.** Treat topology/dynamics only as consciousness-state correlates and require independent state labels.
- **P-17 - stronger partial.** Bifurcation is rejected and path dependence passes in a bounded coupling sweep. Full multi-control phase and autonomous return tests remain.

Reading residuals

- **P-18-P-20 - first tranche complete, named remainder open.** Dretske, Millikan, Boyd, Dennett, and Metzinger are operationalized. Papineau, Swampman/function indeterminacy, realist/pragmatist Boyd critiques, Hohwy, Seth, and the named paper revisions remain.
- **P-21-P-23 - open.** Normativity/ethical thresholds; induction and symmetry choice; consciousness-boundary reading. These must remain separate from mechanism promotion.

Governance, framework, and direction residuals

- **P-24 - stronger partial.** The primer recenters identity around representational reality, but the same firewall is not propagated across the flagship portfolio.
- **P-25 - open.** Add an independent intervention adjudicator with blinded claim/evidence bundles.
- **P-26 - stronger partial.** Twelve machine-readable claims and bidirectional evidence validation exist; abstract-level and flagship coverage remain.
- **P-27 - stronger partial.** Manifests, gate templates, validators, and root CI exist; the repo-local scientific-experiment skill and legacy risk-detection exercise remain.
- **P-29 - stronger partial.** Metadata and provenance freshness checks exist; prose-to-registry staleness and deliberate stale-claim fixtures do not.
- **P-30-P-32 - open programs.** Endogenous stakes is the umbrella for P-10; gauge pluralism must distinguish operational decompositions; consciousness and ethics remain independent precautionary tracks.

Article 4 - Science of the program residuals

Article correction residuals

- **S-002 - open.** Replace universal method claims with generated coverage counts and explicit exceptions.
- **S-003 - stronger partial.** The article distinguishes dispatch from reproduction, provenance has non-mutating freshness, and the root quality environment is locked. Experiment-output verification and clean-clone equivalence remain.
- **S-004 - stronger partial.** Nosek, Efron, Gelman and Carlin, Many Analysts, Belinda Mo, and Bai et al. are connected to repository gates. A complete bibliography and orphan-citation audit remain.
- **S-005 - open.** Replace bare AUC with metric family, positive class, averaging, uncertainty, and decision threshold.
- **S-006 - stronger partial.** The primer now names the independent sampling unit and rejects universal seed floors. Generic preregistration enforcement and empirical calibration remain.
- **S-007 and S-008 - open.** Generate a live unverified panel and repair the split tables.

Replication and correction residuals

- **S-009 - open.** Audit every three-seed claim and rerun only promotion candidates under a declared sampling unit.
- **S-010-S-017 - open independent checks.** Reward-deformation exponent, failed topology mediation, wrong-sign semantic geometry, external weakness kill, uncertainty planner loss, factorial retraction, wide-CI positive, and failed-original-gate near miss.
- **S-018 - open.** Independently replicate a balanced positive set only after row evidence and exact lineage are public.

New experiment residuals

- **S-019 - open.** Independent adversarial-control challenge.
- **S-020 - stronger partial.** M5 demonstrates one locked suite with pre-outcome repair and frozen integrity; it does not test an unseen independent implementation.
- **S-021 - open.** Outsider-authored environments as the external-validity benchmark.
- **S-022 - synthetic tranche complete, empirical residual open.** The 60-cell simulation rejects universal seed floors. Empirical variance structures, BCa/bootstrap-t/randomization comparisons, and preregistered conditional Type-S/Type-M remain.
- **S-023-S-028 - open.** Inject oracle leakage; benchmark experiment invention versus supplied probes; unseen transfer ladder; evidence-dependent mechanism traces; blind claim-tier regrading; auditor blind-spot comparison.
- **S-029 - stronger partial.** M5 supplies one severe computation/integrity audit and cross-platform hash repair. Independent row-level recomputation of flagship negatives is still absent.
- **S-030 - open.** Run one prospective claim family with multiplicity and sequential-stopping control.

Reading, infrastructure, and governance residuals

- **S-031-S-034 - first tranche partial.** Remaining sources include Kerr, Gelman/Loken, selective inference and FDR, Tibshirani, Lakens, adversarial-collaboration protocols, Lu et al., the named replication study, and benchmark/reproduction evidence.
- **S-035 - open.** Ground retrieval/search/discovery in novelty and mechanism-validation research.
- **S-036 and S-038 - stronger partial.** Twelve evidence and twelve claim records, schemas, referential validation, one E5 verdict, and root enforcement exist. Append-only/full-program coverage, threshold expressions, metric/resampling typing, generated reports, and migration remain.
- **S-037 - open.** Claim-family multiplicity and sequential-stopping evaluation.
- **S-039 - open.** Public-safe row evidence for flagship suites.
- **S-040 - stronger partial.** Five experiment manifests plus M5 integrity bindings exist; one immutable execution manifest per run with environment/executor/parent lineage remains.
- **S-041 - open.** Typed metric registry.
- **S-042-S-044 - stronger partial.** Claim tiers/statuses are checked and M5 preserves supersessions and preregistration chronology. Generic maximum-tier rules, immutable transition tests, amendment validation, and stopping-rule checks remain.
- **S-045-S-046 - open.** Reproduction verifier and independent audit bundles.
- **S-047 - stronger partial.** M5 has a complete prose discovery audit; normalized, round-trippable records remain.
- **S-048 - stronger partial.** Publication guard remains mandatory; standardized allowlisted rows, archives, hashes, and policy extensions remain.
- **S-049-S-053 - open directions.** Independence-first governance; external promotion currency; synthetic-world claim ceilings; cumulative evidence graph; discovery as new executable capability plus external verification.

Article 5 - Software engineering residuals

Article correction residuals

- **E-SE-002 - partial.** Correct the architecture map for the Inquiry app, sites, coherence testbench, and generated counts.
- **E-SE-003 - stale partial.** Replace "no root lock" with the actual split: locked root quality group, app lock, E5/E6 locks, and independent experiment/Modal runtime definitions.
- **E-SE-004 - partial.** Replace raw-data absolutes with a typed information-flow map covering private raw, reduced public rows, summaries, paper-primary packages, and exceptions.
- **E-SE-005 - stronger partial.** M5 exposed branch-dependent RNG and sub-scientific cross-platform float differences. Publish explicit algorithmic, semantic, environment, and byte-replay tiers.
- **E-SE-006 - stale partial.** Root pytest coverage is fixed; inventory the excluded Inquiry, coherence, Haskell, site, Modal, and static-analysis surfaces instead of repeating the old 13-file claim.
- **E-SE-007-E-SE-009 - open or partial.** Publish current provenance/reproduction completeness, explain enforcement levels, and add a commit-bound generated status appendix.

Corrective engineering experiments

- **E-SE-010 - open.** Remove hardcoded flagship values and bind every figure/table to committed evidence.
- **E-SE-011 and E-SE-021 - stronger partial.** M5 gives one byte-stable same-platform replay and one macOS/Linux semantic-hash incident. The representative multi-experiment and multi-platform determinism matrix remains.
- **E-SE-012-E-SE-015 - open.** Dependency-drift replication, clean paper rebuild registry, outside flagship capsule, and labeled provenance audit dataset.
- **E-SE-016 - stronger partial.** Command-order, one-environment, bounded-worker, and fail-fast tests exist. Injected secret, syntax, lint, type, pytest, provenance, and paper-drift defects remain.
- **E-SE-017-E-SE-022 - open or partial.** Exporter mutation, Modal chaos/resume, public-row threat modeling, clean-clone regen drill, platform determinism, and negative anti-leak type/import fixtures.

Reading residuals

- **E-SE-023R - stronger partial.** Official uv/setup-uv design informed the landed quality gate. Nix/Guix, Reproducible Builds, and PyTorch determinism synthesis remain.
- **E-SE-024R-E-SE-028R - open.** W3C PROV/RO-Crate/CodeMeta/ACM artifacts; Datasheets/Model Cards/privacy; leases/Chubby/fencing/TLA+; property/schema/type-boundary limits; SLSA/SBOM/Sigstore/OpenSSF/NIST SSDF.

Framework residuals

- **E-SE-023 - stronger partial.** Required root PR CI is real and locked. Add owned lanes or explicit exceptions for Inquiry, coherence, Haskell, sites, and paper smoke; prove branch protection and path-filter resistance.
- **E-SE-024 and E-SE-025 - open or partial.** Fail-closed confirmatory Haskell parity and a declarative evidence-linked paper renderer.

- **E-SE-026 - stronger partial.** Root-quality locking is complete. Runtime dependency registry, approved exceptions, update policy, SBOM, drift check, and Modal/reproduction adoption remain.
- **E-SE-027 - open.** Small installable research-instrument library for namespaced seeds, atomic writes, digests, fingerprints, and safe summaries.
- **E-SE-028 and E-SE-029 - stronger partial.** Gate schemas/validators/templates and five manifests exist. PASS/inconclusive migrations, scaffolder, complete package migration, and owner/reason/expiry exceptions remain.
- **E-SE-030-E-SE-034 - partial.** Shared public envelope, Modal orchestration/maturity registry, typed launch surfaces, structured provenance v2, and hardened publication scanning remain incomplete.
- **E-SE-035-E-SE-042 - open.** Repo-local science-experiment skill; host-agnostic config; generated inventory; versioned instrument kernel; evidence capsule tiers; claim graph; risk budget; formal trust-infrastructure state machines.

Article 6 - Systems theory and complexity residuals

Article correction residuals

- **T-SYS-001 - partial.** Main title metadata is repaired; exact full title/subject and deterministic identity gate remain.
- **T-SYS-002 - stronger partial.** The primer and autopoietic paper are bounded, but downstream passive-to-active writing still uses unqualified active-attractor language.
- **T-SYS-003 - open.** Repair pagination, widows, and orphaned callouts.
- **T-SYS-004 - stronger partial.** One coupling/order-parameter case exists; the comprehensive control/order/null/claim-tier map does not.

Old experiment residuals

- **T-SYS-005 - open.** Reanalyze abrupt load-bearing against continuous and thresholded nulls.
- **T-SYS-006 - open.** Add autonomous state dynamics, return maps, basin geometry, perturbation recovery, and stability eigenvalues.
- **T-SYS-007-T-SYS-010 - open.** Quantitative Stack law, complete dimension sweep, repaired self-repair controls, and supervised-versus-experience-derived geometry consolidation.

New experiment residuals

- **T-SYS-013 - open.** Multi-control phase diagram after the one-dimensional coupling study.
- **T-SYS-014-T-SYS-016 - open.** Many-part virtual governor, collective navigation/topology, and populations of encoders under viability selection.
- **T-SYS-017 - open.** Strict component and boundary production rather than supervised repair.
- **T-SYS-018 and T-SYS-019 - open.** Formal viability kernels, throughput, and maintenance cost.

Reading, framework, and direction residuals

- **T-SYS-020-T-SYS-027 - open reading program.** Transition methods; self-organization versus supervision; distributed coordination; autopoiesis/closure/viability; hierarchy/near-decomposability/Stack theorem; rate-distortion; dissipative structures; no-hype criticality.
- **T-SYS-028 - stronger partial.** A package-local deterministic phase-map runner, clustered bootstrap, controls, and tests exist. A reusable framework and analytic bifurcation fixtures remain.
- **T-SYS-029 and T-SYS-030 - open.** Reproducible many-agent environment and dynamical basin/stability utilities.
- **T-SYS-031 - stronger partial.** Claims and evidence fail closed for selected studies; a seven-term complexity-vocabulary lint and bad-fixture suite remain.
- **T-SYS-032-T-SYS-034 - open or partial directions.** Explicit system tiers; separate thermodynamic self-maintenance track; boundary production and closure graph as the decisive autopoiesis frontier.

Recommended immediate queue

- 1 **Residual ledger sync.** Update the six backlogs and two affected primers to current main. This is documentation truth repair, not new science.
- 2 **Program contract migration.** Add M5 canonical evidence/claim/gate records, then migrate a PASS and an inconclusive result. Define immutable status and preregistration-amendment checks.
- 3 **Public evidence pilot.** Choose one positive and one negative flagship result; publish minimum safe rows and exact computation manifests; make regen verify reproduce the reported statistics.
- 4 **Prospective statistics pilot.** Register one claim family with sampling units, multiplicity, sequential stopping, Type-S/Type-M targets, and promotion rules.
- 5 **Independent audit.** Give the bundles to an outside implementation/reviewer before launching another large confirmatory sweep.
- 6 **Decisive science pair.** Run M-201 theorem-necessity completion and P-10 endogenous-concern controls in parallel; both test central claims without requiring the largest external compute spend.
- 7 **Dynamics and external contact.** Only after the trust substrate and independent audit pass, run T-SYS-006 return/basin diagnostics and one outsider-authored external-validity benchmark.

Guardrails for closing residuals

- A negative or failed gate closes an experiment if the frozen test was executed correctly; it does not validate the positive hypothesis.
- An invalid integrity gate closes neither side of a scientific claim.
- A synthetic result may close a synthetic tranche while leaving empirical transport open.
- A reading task closes only when its sources change a claim boundary, control, estimator, schema, or decision rule.
- No item closes from prose alone when its gate asks for executable evidence.
- Do not reopen a completed negative by tuning thresholds or controls against its outcomes. A replacement requires a new question and preregistration.
- Treat current manifests, registries, and root CI as a foundation to migrate and extend, not as absent infrastructure and not as universal coverage.

Source record

- Six article-specific residual sources under docs/primers/backlogs/.
- Active ledger: TODO.md at origin/main commit 0413098.
- Discovery audit: docs/discovery_regime_audit.md.
- M5 result: experiments/commitment_surface/results/m5_suite_c_reopen_reset_trigger_2026_07_14.md.
- Root quality implementation: scripts/run_quality_checks.py, .github/workflows/quality.yml, pyproject.toml, and uv.lock.
- Primer implementation commits merged through PR #357, M5 through PR #359, and locked quality work through PR #361.